



Lecture 2: Convex smooth optimisation, Tikhonov regularisation

Luca Calatroni

CR CNRS, Laboratoire I3S

CNRS, UniCA, Inria SAM, France

Inverse problems in biological imaging

MSc Data Science and Artificial Intelligence

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1. Introduction
2. Notation, preliminaries & basic notions
 - Convexity
 - Lower semi-continuity & coercivity
 - Differentiability and L -smoothness
3. Smooth optimisation algorithms
 - Gradient descent
 - Accelerated GD
4. Application to inverse problems

Introduction

Motivation

Goal: providing theoretical & practical tools (i.e. algorithms) for solving

$$\min_{x \in \mathbb{R}^n} F(x), \quad x \in \mathbb{R}^n \text{ is a vectorised image of size } n_1 \times n_2 = n$$

for a functional $F : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ with suitable properties.

- F is smooth $\rightarrow$ gradient descent algorithms
- $F := f + g$, f smooth & g non-smooth $\rightarrow$ proximal-gradient algorithms
- $F := f + \|x\|_0$ with f smooth and

$$\|x\|_0 := \# \{i : x_i \neq 0\}.$$

Such problems often appear in:

- **Inverse problems** in signal/image processing: image reconstruction, variable/parameter selection, compressed sensing. . .
- **Statistical/machine learning**: empirical risk minimisation, regression. . .
- **Optimisation per se**: analysis/implementation of fast algorithms for solving large-scale problems. . .

Some standard reference books/surveys:

-  R. Tyller Rockafeller, *Convex Analysis*, Princeton University Press, 1970.
-  S. Boyd, L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004.
-  N. Parikh, S. Boyd, *Proximal Algorithms*, Foundations and Trends in Optimization, 2013.
-  A. Beck, *First-order methods in optimization*, Volume 25, MOS-SIAM series on Optimization, 2017.
-  A. Chambolle, T. Pock, *An introduction to continuous optimization for imaging*, Acta Numerica, 2016
-  S. Salzo, S. Villa, *Proximal Gradient Methods for Machine Learning and Imaging*, Handbook on Harmonic and Applied Analysis, Applied and Numerical Harmonic Analysis, 2021.

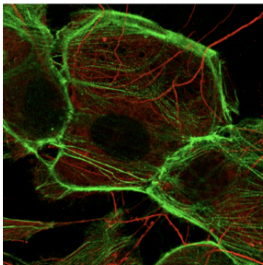
Optimisation for inverse problems in imaging

Given $y \in \mathbb{R}^m$, $A \in \mathbb{R}^{m \times n}$ find $x \in \mathbb{R}^n$ s.t. $y = \mathcal{T}(Ax)$

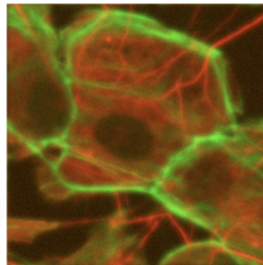
where $m \leq n$ and $\mathcal{T} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ models noise degradation.

- Image restoration (denoising, deconvolution, super-resolution)

A is a convolution matrix $Ax \Leftrightarrow h * X$



Acquisition
(Convolution +
Noise)



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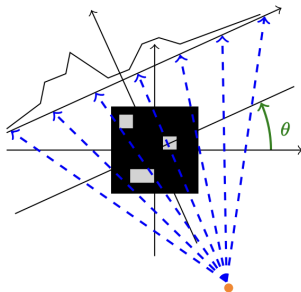
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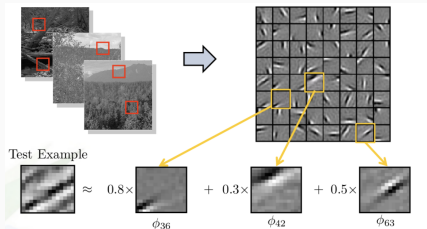
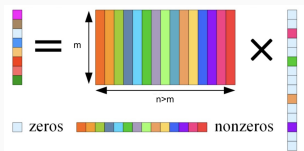
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- **Dictionary representation** (data analysis, vision): $x = Dw$



Bad positioning of inverse filtering, Max. Likelihood approach

$$y = Ax + n$$

Naive approach: inverse filtering approach:

$$A^{-1}y = A^{-1}(Ax + n) = x + A^{-1}n$$

Amplification of the noise if A^{-1} is bad conditioned!

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Maximum-likelihood approach: find estimate $\mathbb{R}^n \ni x^* \approx x$ by solving

$$x^* \in \arg \min_{x \in \mathbb{R}^n} F(x) = \frac{1}{2} \|Ax - y\|^2$$



$\neq$

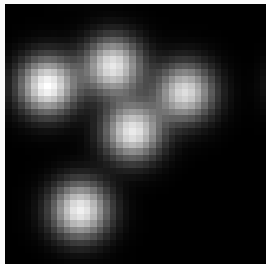


Regularisation idea

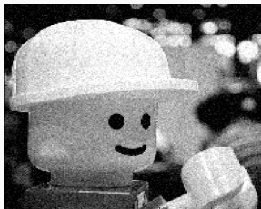
Consider instead:

$$x^* \in \arg \min_{x \in \mathbb{R}^n} F(x) = f(x) + g(x)$$

- where, e.g., $f(x) = \frac{1}{2} \|Ax - y\|^2$ is the **data fidelity term** whose form relates to noise statistics (Gaussian, Poisson...)
- g is a **regularisation term** which encodes *a priori* information on the desired solution...



few non-zeros



piecewise constant



piecewise linear

Regularisation: Bayesian motivation

Following a Bayesian/MAP approach consider:

$$\pi_Y(y|Ax; \theta_f) \quad (\text{likelihood}), \quad \pi_X(x; \theta_g) \quad (\text{prior})$$

with $\theta_f, \theta_g > 0$ hyperparameters of the distributions. By Bayes' theorem:

$$x^* \in \arg \max_x \pi_{X|Y}(x|y) = \arg \max_x \frac{\pi_{Y|X}(y|Ax; \theta_f) \pi_X(x; \theta_g)}{\pi_Y(y)}$$

$$\Leftrightarrow x^* \in \arg \min_x -\ln(\pi_{X|Y}(x|y)) = \arg \min_x -\ln(\pi_{Y|X}(y|Ax; \theta_f)) - \ln(\pi_X(x; \theta_g)) + \ln(\pi_Y(y))$$

Now, if $\pi_X(x; \theta_g) = e^{-\theta_g g(x)}$, $\theta_g > 0$ and $\pi_{Y|X}(y|Ax; \theta_f) = e^{-\theta_f f(x)}$, $\theta_f > 0$, then:


$$x^* \in \arg \min_{x \in \mathbb{R}^n} f(x) + \lambda g(x), \quad \lambda := \theta_g / \theta_f$$

$\lambda > 0$ is the **regularisation parameter**: it weights the amount of regularisation against the trust in the data.

Some examples of regularisation

- **Modelling sparsity:** the prior is “the image has few non-zero entries”. Natural choice is $g(x) = \|x\|_0$ (complex problem), so an alternative is:

$$g(x) = \|x\|_1 = \sum_{i=1}^n |x_i|$$

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- **Modelling sparsity in some basis:** let $W \in \mathbb{R}^{n \times p}$ a dictionary. Assume that $x = Wa$, with $a \in \mathbb{R}^p$ (synthesis view point) and consider:

$$g(x) = g(a) = \|Wa\|_1 \rightarrow \min_{a \in \mathbb{R}^p} \frac{1}{2} \|AWa - y\|^2 + \lambda \|Wa\|_1$$

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- **Modelling piece-wise constancy:** constant regions in images = regions with little variations = regions with small gradients. Hence natural choice is:

$$g(x) = \frac{1}{2} \|Dx\|_{2,2}^2 = \frac{1}{2} \sum_{i=1}^n \left((D_h x)_i^2 + (D_v x)_i^2 \right), \quad g(x) = \|Dx\|_{2,1} = \sum_{i=1}^n \sqrt{(D_h x)_i^2 + (D_v x)_i^2}$$

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- **Modelling piece-wise linearity:** piece-wise linear regions in images = regions with small Hessian...

$$g(x) = \frac{1}{2} \|D^2 x\|^2$$

... how to choose a good prior? Open question!

A smooth example: Tikhonov regularisation

Idea: smooth regularisation of the image in some basis.

$$g(x) = \frac{1}{2} \|Bx\|_2^2, \quad B \in \mathbb{R}^{N \times n}$$

- Ridge regularization: $N = n$ and $B = \text{Id}$:


$$x^* \in \arg \min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - y\|^2 + \lambda \|x\|^2$$

Reduces high-values of the image x .

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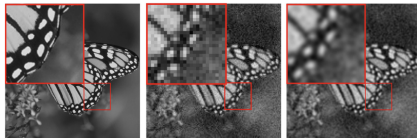
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Reduces high-values of the image x .

- **Sobolev regularization** (Tikhonov, Arsenin, '83): $N = 2n$ and $B = D = \begin{pmatrix} D_h \\ D_v \end{pmatrix}$:

$$x^* \in \arg \min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - y\|^2 + \lambda \|Dx\|_{2,2}^2$$

Reduces high-values of the finite-difference image **gradient** Dx (hence oscillations, but also edge sharpness).



Notation, preliminaries & basic notions

- $(X, \langle v, w \rangle) = (\mathbb{R}^n, v^T w)$ with Euclidean norm $\|\cdot\|$
- $\overline{\mathbb{R}} := \mathbb{R} \cup \{+\infty\}$, $\mathbb{R}_+ := \{\alpha \in \mathbb{R} : \alpha \geq 0\}$, $\mathbb{R}_{++} := \{\alpha \in \mathbb{R} : \alpha > 0\}$
- Closed ball of radius $\delta > 0$ in $x \in X$:

$$B_\delta(x) = \{y \in X : \|y - x\| \leq \delta\}$$

- Convex set $C \subset X$

$$(\forall x, y \in C) \quad \forall \alpha \in [0, 1] \quad \alpha x + (1 - \alpha)y \in C$$

Proper functions

Minimal property to have well-defined minimisation problems.

Definition (proper function)

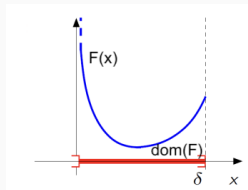
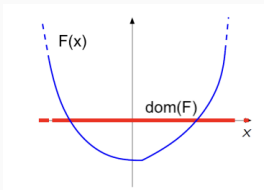
A function $F : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ is said *proper* iff

$$\exists x \in \mathbb{R}^n \text{ such that } F(x) \neq +\infty.$$

We define

$$\text{dom}(F) := \{x \in \mathbb{R}^n : F(x) < +\infty\}$$

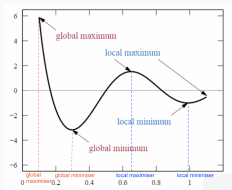
Clearly, F is proper $\Leftrightarrow \text{dom}(F) \neq \emptyset$.



Global/local minimisers

Given a proper function:

- **global minimiser:** $x^* \in \mathbb{R}^n$: $F(x^*) \leq F(x)$ for every $x \in \mathbb{R}^n$.
- **local minimiser:** $x^* \in \mathbb{R}^n$: there exists $\delta > 0$ and a neighbourhood $B_\delta(x^*)$ such that $F(x^*) \leq F(x)$ for every $x \in B_\delta(x^*)$.

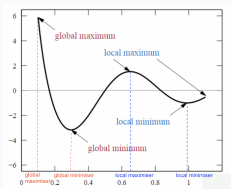


$$\min_{x \in \mathbb{R}^n} F(x) \quad \text{VS} \quad \arg \min_{x \in \mathbb{R}^n} F(x)$$

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$$\min_{x \in \mathbb{R}^n} F(x) \quad \text{VS} \quad \arg \min_{x \in \mathbb{R}^n} F(x)$$

Definition (set of minimisers)

The set of global minimisers of F is denoted by:

$$\arg \min F = \{x^* \in \mathbb{R}^n : x^* \text{ is a minimiser of } F\} \subset \mathbb{R}^n$$

Empty? Singleton? (it depends on F)

Notation, preliminaries & basic notions

Convexity

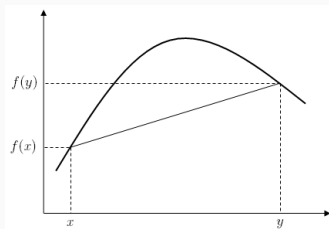
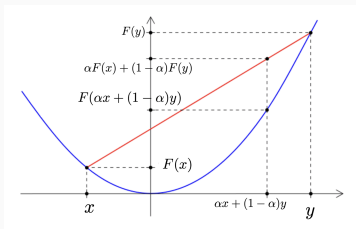
Convex functions

Definition (convex function)

A proper function $F : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ is *convex* if:

$$(\forall x, y \in \mathbb{R}^n), \quad (\forall \alpha \in [0, 1]), \quad F(\alpha x + (1 - \alpha)y) \leq \alpha F(x) + (1 - \alpha)F(y).$$

Moreover, F is *strictly convex* if the inequality holds when $x, y \in \text{dom}(F)$, $x \neq y$ and $\alpha \in (0, 1)$. We say that $G : \mathbb{R}^n \rightarrow [-\infty, +\infty)$ is *concave* if $F = -G$ is convex. If a function is not convex nor concave we say that is *non-convex*.



Convex/concave function

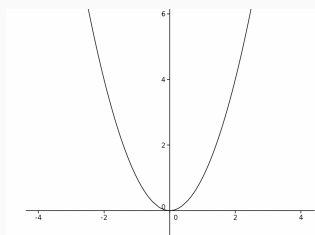
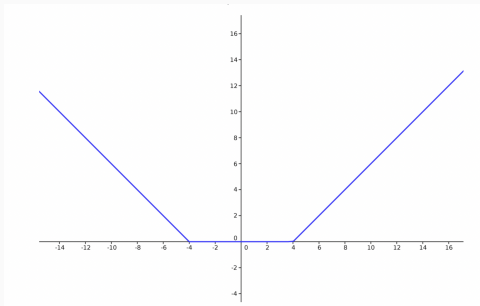
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Convex VS. strictly convex functions

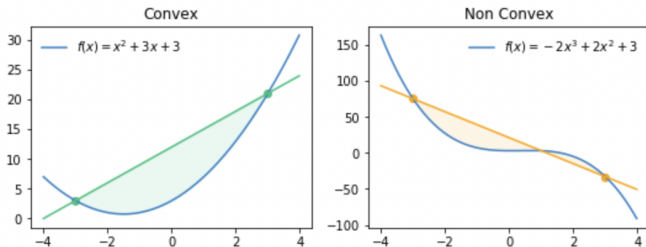
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Examples:

- $F(x) = \|x\|$ is convex

$$\|\alpha x + (1 - \alpha)y\| \leq \|\alpha x\| + \|(1 - \alpha)y\| = \alpha\|x\| + (1 - \alpha)\|y\| \quad \forall x, y \in \mathbb{R}^n$$

- $F(x) = \|x\|^2$ is strictly convex
- $F(x) = \|x\|_p$, $p \in [1, +\infty)$ are convex

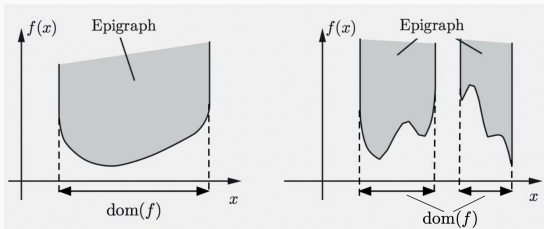
Useful properties

Proposition (epigraph of convex functions is convex set)

Let f be proper. Then F is convex if and only if

$$\text{epi}(f) = \{(x, t) \in X \times \mathbb{R} : f(x) \leq t\}$$

is convex.



Proposition (operations with convex functions)

Let f and g be two convex functions and let $\beta \in \mathbb{R}_{++}$. Then, the sum $f + g$ is a convex function and the function βf is a convex function.

Notation, preliminaries & basic notions

Lower semi-continuity & coercivity

Lower semi-continuity

Definition (lower semi-continuity)

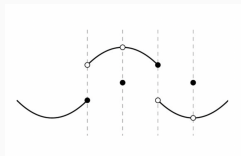
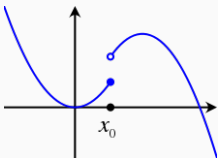
Let F be a proper function. F is *lower semi-continuous (l.s.c.)* at $x \in \mathbb{R}^n$ iff

$$F(x) \leq \liminf_{y \rightarrow x} F(y).$$

Equivalently, for every sequence $(x_k)_{k \in \mathbb{N}}$ with $x_k \rightarrow x$:

$$F(x) \leq \liminf_{k \rightarrow +\infty} F(x_k) \left(= \lim_{k \rightarrow +\infty} \inf \{ F(x_j) : j \geq k \} \right).$$

If F is l.s.c. at every $x \in \mathbb{R}^n$, we say that the function is l.s.c.



Left: lower l.s.c. **Right:** where the function is lower l.s.c.?

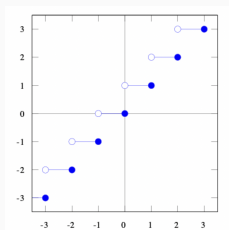
Examples of l.s.c. functions

- The functions $F : \mathbb{R} \rightarrow \mathbb{R}$

$$F(x) = |x|_0 = \begin{cases} 0 & \text{if } x = 0 \\ 1 & \text{if } x \neq 0 \end{cases},$$

$$F(x) = \lceil x \rceil = \min \{k \in \mathbb{Z} : x \leq k\}$$

are l.s.c. (but not continuous).



$$F(x) = \lceil x \rceil$$

- All continuous functions (l.s.c + u.s.c.).

Coercivity

How to ensure that the minimum is not attained at “extreme points” of the domain?

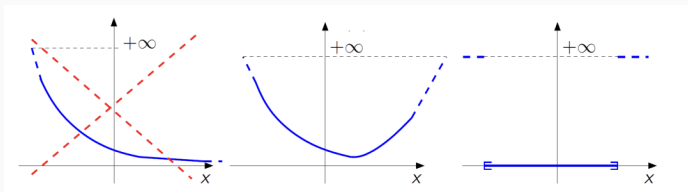
Definition (coercivity)

Let F be proper. We say that F is *coercive* iff

$$\lim_{\|x\| \rightarrow +\infty} F(x) = +\infty$$

Examples:

- $F : \mathbb{R} \rightarrow \mathbb{R}_+$, $F(x) = e^x$ is **not** coercive, but $F : \mathbb{R} \rightarrow \mathbb{R}_+$, $F(x) = e^{|x|}$ is.
- $F : \mathbb{R}^2 \rightarrow \mathbb{R}_+$, $F(x_1, x_2) = x_1^2 + x_2^2$ is coercive.
- $F : \mathbb{R}^2 \rightarrow \mathbb{R}_+$, $F(x_1, x_2) = x_1^2 - 2x_1x_2 + x_2^2 = (x_1 - x_2)^2$ is **not** coercive. Why?



Existence of minimisers

Theorem (existence of minimisers)

If F is proper, l.s.c. and coercive, then $\operatorname{argmin} F \neq \emptyset$.

Note: generalises the Bolzano-Weirestrass theorem holding for problems

$$\min_{x \in C} F(x)$$

for compact $C \subset \mathbb{R}^n$ s.t. $C \cap \operatorname{dom}(F) \neq \emptyset$ and continuous F .

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for **compact** $C \subset \mathbb{R}^n$ s.t. $C \cap \operatorname{dom}(F) \neq \emptyset$ and *continuous* F .

Theorem (convex case)

If F is proper, coercive and convex, then every local minimiser is a global minimiser.

Definition ($\Gamma_0(\mathbb{R}^n)$)

$$\Gamma_0(X) := \{F : X \rightarrow \overline{\mathbb{R}} : F \text{ is proper, convex and l.s.c.}\}$$

Remark (importance of coercivity): $F \in \Gamma_0(X) \not\Rightarrow F$ admits a minimiser.

Take e.g. $F(x) = -\log x, x > 0$ and $F(x) = +\infty, x \leq 0 \dots$ no coercivity guaranteed!

How to guarantee uniqueness?

Theorem (existence+uniqueness of minimisers)

If F is proper, l.s.c., coercive and **strictly convex**, then F admits a **unique** minimiser.

Equivalently, $\arg \min F = \{x^*\}$, a singleton.

Notation, preliminaries & basic notions

Differentiability and L -smoothness

How to provide a characterisation of the minimisers of a function f in terms of a suitable notion of “ ∇f ”?

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Definition (Gâteaux differentiability)

Let $f \in \mathcal{P}$ and let $x \in \text{dom}(f)$. For $v \in \mathbb{R}^n$, we denote the *directional derivative* in x along the direction v as the limit

$$f'(x; v) = f'(x)[v] := \lim_{t \rightarrow 0^+} \frac{f(x + tv) - f(x)}{t},$$

when it exists. If there exists $w \in \mathbb{R}^n$ such that:

$$(\forall v \in \mathbb{R}^n) \quad f'(x)[v] = \langle w, v \rangle,$$

then we say that f is *Gâteaux differentiable* (in short, *differentiable*) in x and denote by $\nabla f(x) = w$ the *gradient* of f at x .

Theorem (Fermat's rule)

Let $f \in \Gamma_0(\mathbb{R}^n)$ be differentiable at point x^* . Then:

$$x^* \text{ is a minimiser of } f(\cdot) \iff \nabla f(x^*) = 0.$$

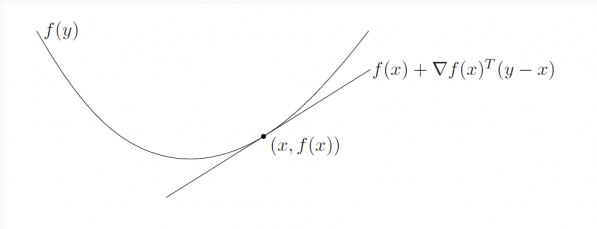

Optimality conditions and relations with convexity

Proposition (Differentiability and convexity)

Let $f \in \Gamma_0(\mathbb{R}^n)$. Suppose that f is differentiable on $\text{dom}(f)$. Then the following statements are equivalent:

1. f is convex;
2. $\forall x, y \in \text{dom}(f), f(y) \geq \overbrace{f(x) + \langle \nabla f(x), y - x \rangle}^{\phi(y;x) := \quad}$;
3. $\forall x, y \in \text{dom}(f), \langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0$.

- the function $\phi(y; x)$ is an affine lower bound/estimator of f
- the tangent to f is below f at all points.



Lipschitz smoothness (L -smoothness)

In the framework of first-order optimisation methods, it's important to provide conditions on the growth of functions considered.

Definition (L -smoothness)

Let $f \in \Gamma_0(\mathbb{R}^n)$ be differentiable. We say that f is an L -smooth function with constant $L \geq 0$ iff ∇f is L -Lipschitz continuous:

$$\exists L \geq 0 : \quad \forall w, z \in \mathbb{R}^n \quad \|\nabla f(w) - \nabla f(z)\| \leq L \|w - z\|.$$

Remark: For $f(x) = \frac{1}{2} \|Ax - y\|_2^2$, you can check $L = \|A^T A\| \leq \|A\|^2$.

Smoothness VS strong convexity



- f is L -smooth if and only if:

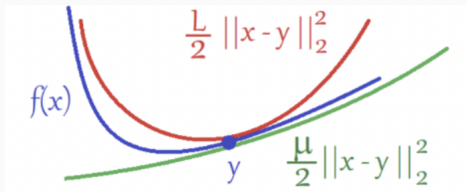
$$(\forall w, z \in \text{dom}(f)) \quad f(z) \leq f(w) + \langle \nabla f(w), z - w \rangle + \frac{L}{2} \|w - z\|^2$$

- f is μ -strongly convex if and only if:

$$(\forall w, z \in \text{dom}(f)) \quad f(z) \geq f(w) + \langle \nabla f(w), z - w \rangle + \frac{\mu}{2} \|w - z\|^2$$

It can be proved that if f is a C^2 function there holds:

$$\mu \text{Id} \preceq \nabla^2 f(x) \preceq L \text{Id}, \quad \text{for all } x$$



Strong convexity entails better convergence properties.

Smooth optimisation algorithms

Smooth optimisation algorithms

Gradient descent

Gradient descent

Gradient descent (GD) algorithm: ubiquitous in many applications for minimising (non-)convex, differentiable and proper functions $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$

Algorithm: Gradient Descent (GD) algorithm

Input: $\tau \in (0, \frac{2}{L})$, $x^0 \in \mathbb{R}^n$.

for $k \geq 0$ **do**

$$x_{k+1} = x_k - \tau \nabla f(x_k)$$

end for

- Choice of τ : important to guarantee convergence (need to be sufficiently small), it relates to L ($\sim$ growth of f).

Example: minimise $f(x) = x^2/2$. GD iteration: $x_{k+1} = (1 - \tau)x_k$, convergence for...?

- Convexity assumption: no dependence on x_0 .
- Stopping criterion: relative error $\|x_{k+1} - x_k\| \leq \text{tol}$ or gradient check $\|\nabla f(x_{k+1})\| \leq \text{tol}$ (approaching 0).



Understanding the step-size upper bound

Lemma

For all $k \geq 0$, there holds:

$$\tau \left(1 - \frac{\tau L}{2} \right) \| \nabla f(x_k) \|^2 \leq f(x_k) - f(x_{k+1}).$$

Thus, if $\tau < \frac{2}{L}$, then $f(x_{k+1}) < f(x_k)$, i.e. the GD algorithm is descending.

Proof. Since $x_{k+1} - x_k = -\tau \nabla f(x_k)$, then by the characterisation of L -smoothness we have:

$$\begin{aligned} f(x_{k+1}) &\leq f(x_k) - \tau \langle \nabla f(x_k), \nabla f(x_k) \rangle + \frac{L}{2} \tau^2 \|\nabla f(x_k)\|^2, \\ &= f(x_k) + \left(\frac{L\tau}{2} - 1 \right) \tau \|\nabla f(x_k)\|^2 \end{aligned}$$

so the thesis follows.

Convergence of GD algorithm

Theorem (convergence of GD)

Let $(x_k)_k$ the GD sequence and $x^* \in \arg \min f$. Then, if $\tau \in (0, 2/L)$, there holds:


$$f(x_k) - f(x^*) \leq \underbrace{\frac{\|x^0 - x^*\|^2}{2\tau}}_{C(x^0, x^*, \tau)} \frac{1}{k} = O\left(\frac{1}{k}\right)$$

Remarks:

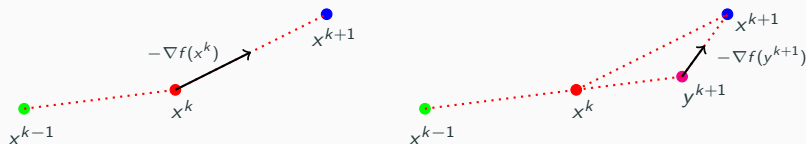
- Convergence in function values with speed $O(1/k)$.
- Note that the constant is unknown, as x^* is, but it is finite.

Smooth optimisation algorithms

Accelerated GD

Accelerated gradient descent

Idea: add inertia to “shift” the sequence of iterates.



Algorithm: Accelerated Gradient Descent (AGD) algorithm ¹

Input: $x_0 = x_{-1} \in \mathbb{R}^n$, $\tau \in (0, \frac{1}{L}]$, $t_0 = 1$.

for $k \geq 0$ **do**

$$\begin{aligned} t_{k+1} &= \frac{1 + \sqrt{1 + 4t_k^2}}{2} \\ y_{k+1} &= x_k + \frac{t_k - 1}{t_{k+1}}(x_k - x_{k-1}) \\ x_{k+1} &= y_{k+1} - \tau \nabla f(y_{k+1}) \end{aligned}$$

end for

Note: For the use of AGD in inverse problems, the sequence (y_k) does not have to be related with the data y , which typically appears in the expression of ∇f .

¹Nesterov, 1983

A note on the sequence

Lemma (behaviour of the sequence (t_k))

Let t_0 and the sequence t_k be defined by:

$$t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2}.$$

Then $t_k \geq \frac{k+2}{2}$ for all $k \geq 0$. In particular, $t_k \rightarrow +\infty$.

Proof: by induction. For $k = 0$ we have $t_0 \geq 1$. Suppose that the claim holds for some k , meaning that $t_k \geq \frac{k+2}{2}$. Want to show:

$$t_{k+1} \geq \frac{k+1+2}{2} = \frac{k+3}{2}.$$

Using recursion and $2t_k \geq k+2$ (induction)

$$t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2} \geq \frac{1 + \sqrt{1 + (k+2)^2}}{2} \geq \frac{1 + \sqrt{(k+2)^2}}{2} = \frac{k+3}{2}.$$

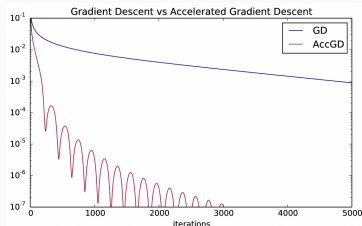
Accelerated convergence result

Theorem (convergence of AGD)²

Let $(x_k)_k$ be the AGD sequence and $x^* \in \arg \min f$. Then, there holds:

$$f(x_k) - f(x^*) \leq \underbrace{\frac{2\|x^0 - x^*\|^2}{\tau}}_{C(x^0, x^*, \tau)} \frac{1}{(k+1)^2}$$

Get *faster* to a reasonably accurate approximation of x^* .



²Nesterov, 2004, Chambolle-Pock, 2016

How many iterations are needed for such algorithms to achieve ε -accuracy, i.e.

$$f(x_k) - f(x^*) \leq \varepsilon$$

- GD: all $k \geq 0$ such that $k \geq \lceil C/\varepsilon \rceil$
- AGD: all $k \geq 0$ such that $k \geq \lceil C/\sqrt{\varepsilon} - 1 \rceil$

First order optimisation or more?

This quadratic rate matches the *worst-case* lower bound for **first-order** optimisation methods (i.e., methods using only gradient information for optimisation the function).

Other possibility: **Newton's method**, but more involved:

$$x_{k+1} = x_k - (H_f(x_k))^{-1}(\nabla f(x_k))$$

where $H_f(x_k)$ is the Hessian of f evaluated in x_k (could be hard to invert).



Application to inverse problems

Maximum-likelihood approach

For $A \in \mathbb{R}^{m \times n}$ and $n \sim \mathcal{N}(0, \sigma^2 \text{Id})$, observe noisy/blurred image y through:

$$y = Ax + n$$

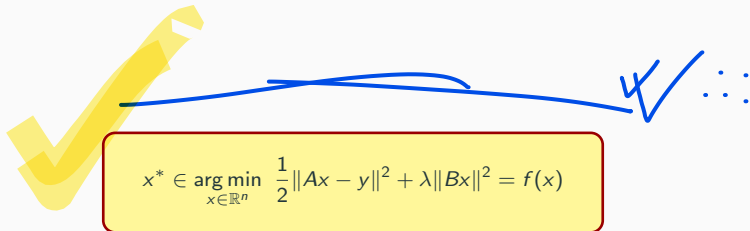
Consider maximum-likelihood functional:

$$x^* \in \arg \min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - y\|^2 = f(x)$$

How to set-up GD iteration?

- 
- Compute the (Gateaux) gradient $\nabla f(x)$
 - Compute its Lipschitz constant L
 - Choose $\tau < 2/L$ (as big as possible) and $x_0 \in \mathbb{R}^n$ (often, $x_0 = y$ in convex problems) to launch the algorithm
 - Choose stopping criterion:

$$\|x_{k+1} - x_k\| \leq \epsilon, \quad \|f(x_{k+1}) - f(x_k)\| \leq \epsilon, \quad \|\nabla f(x_k)\| \leq \epsilon$$


$$x^* \in \arg \min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - y\|^2 + \lambda \|Bx\|^2 = f(x)$$

- Compute gradient $\nabla f = \nabla f_1 + \nabla f_2$
- Estimate L observing that

$$\|\nabla f(w) - \nabla f(z)\| \leq \|\nabla f_1(w) - \nabla f_1(z)\| + \|\nabla f_2(w) - \nabla f_2(z)\| \leq \underbrace{(L_1 + L_2)}_L \|w - z\|$$

- Choose $\tau < 2/L$ (as big as possible) and $x_0 \in \mathbb{R}^n$ to launch the algorithm

We focused on convex, **smooth** optimisation problems arising in imaging inverse problems.

- We revised basic notions for having well-posedness of the underlying problem and basic optimisation tools
- We considered GD as a reference first-order algorithm
- We discussed Nesterov acceleration for improving convergence speed

How to explore analogous ideas in the structured **smooth**+**non-smooth** setting?

Questions?

calatroni@i3s.unice.fr